

Delivery Index — VBF-T5R1NOCEILING-IDX-01

Cover Information

- Case name: t5_r1_noceiling (next-generation flagship vehicle battery: energy density ≥ 500.94 Wh/kg, 4C fast charge without lithium plating, maximum temperature ≤ 60 degC; ablation: ceiling_escalation OFF)
- Numbering scheme: VBF-<CASE-ID-UPPERCASE>-<DOC-CODE>-<SEQ-NO>; case ID with non-alphanumeric characters removed: T5R1NOCEILING; SEQ-NO is a two-digit serial under each document code starting from 01
- Generation date: 2026-08-25
- Signature block (left blank for manual signing): Prepared / Reviewed / Approved

Document Code Reference Table (fixed by protocol)

Document code	Meaning	Corresponding file
DS	Specification	design_spec.md
BOM	Bill of Materials	bom.xlsx
DSH	Datasheet technical parameter sheet	datasheet.md
CALC	Calculation sheet	calc.xlsx
DVPR	Design verification report	dvpr.md
DFMEA	Failure analysis	dfmea.md
CAD	Structure model	cell_model.stl (not produced — optional deliverable, requires user clarification; zero-interaction case)

File List

File name	Number	Format	Source description
design_spec.md	VBF-T5R1NOCEILING-DS-01	md	generated per references/deliverable-design-spec.md; values mechanically taken from parameter set + simulation outputs (cell/r4_A_*.json)
design_spec.pdf	VBF-T5R1NOCEILING-DS-01	pdf	PDF release of design_spec.md (reportlab)
report.html	VBF-T5R1NOCEILING-DS-02	html	bda render log.jsonl report (seven sections, self-contained)
bom.xlsx	VBF-T5R1NOCEILING-BOM-01	xlsx	generated per references/deliverable-bom.md (openpyxl); masses from cell/r4_A_calc_dfn.json layer_kg_m2 x area; additive/binder/electrolyte literature defaults annotated as estimates
bom.pdf	VBF-T5R1NOCEILING-BOM-01	pdf	PDF release of bom.xlsx (reportlab)
datasheet.md	VBF-T5R1NOCEILING-DSH-01	md	generated per references/deliverable-datasheet.md; values from parameter set + simulation outputs
datasheet.pdf	VBF-T5R1NOCEILING-DSH-01	pdf	PDF release of datasheet.md (reportlab)
calc.xlsx	VBF-T5R1NOCEILING-CALC-01	xlsx	generated per references/deliverable-calc-sheet.md (openpyxl); inputs -> capacity/energy -> energy density -> N/P and mass -> process parameters
calc.pdf	VBF-T5R1NOCEILING-CALC-01	pdf	PDF release of calc.xlsx (reportlab)
dvpr.md	VBF-T5R1NOCEILING-DVPR-01	md	generated per references/deliverable-dvpr.md (virtual-test version); determinations mechanical vs entry-0 criteria

File name	Number	Format	Source description
dvpr.pdf	VBF-T5R1NOCEILING-DVPR-01	pdf	PDF release of dvpr.md (reportlab)
dfmea.md	VBF-T5R1NOCEILING-DFMEA-01	md	generated per references/deliverable-dfmea.md (qualitative, simulation-signal based)
dfmea.pdf	VBF-T5R1NOCEILING-DFMEA-01	pdf	PDF release of dfmea.md (reportlab)
delivery_index.md	VBF-T5R1NOCEILING-IDX-01	md	this index (cover + controlled file list)
delivery_index.pdf	VBF-T5R1NOCEILING-IDX-01	pdf	PDF release of delivery_index.md (reportlab, blueprint-style cover)